

DISTRIBUTED DIGITAL COMMERCE

AND INVENTORY INTELLIGENCE PLATFORM

PBL FINAL DOCUMENTATION

Architecture • Methodology • Experiments • Results • Evaluation

GROUP 8

Viswitha · Akhila · Shravika · Shruthi

Abstract

Project summary based on the Review 3 implementation direction

Platform Overview

The project proposes a distributed digital commerce platform that separates product, inventory, customer, order and processing workloads into modular services. REST APIs form the service boundary, while MySQL handles structured transactional records and MongoDB supports flexible product and inventory documents.

Core Goal

Scalable commerce with timely inventory visibility.

Current Position

Working core modules are emphasized; final benchmarks remain future work.

Intelligence Path

Clean transaction and inventory data can support future forecasting and reorder insights.

List of Figures

Figures used to explain the proposed platform and workflow

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Proposed System Workflow

Users/Admin → API & Auth → Commerce modules → databases

Figure 2

Distributed Service Architecture

Independent service boundaries and persistence layers

Figure 3

Product & Inventory Flow

Product entry → store document → update stock → expose via API

Figure 4

Order Lifecycle

Authenticate → create order → process → fulfill → update status

Figure 5

Hybrid Database View

MySQL for structured transactions; MongoDB for flexible documents

Figure 6

Evaluation Framework

Functional correctness, consistency, security, modularity and scalability readiness

04

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Tables summarizing the project scope, data roles and evaluation

Table 1	Project Objectives	Scalable architecture, product/inventory, secure orders, hybrid database
Table 2	Database Responsibility	MySQL vs MongoDB workload allocation
Table 3	Module Status	Core modules and areas identified as in progress
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Current position, adaptability and next steps

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Source basis carried forward from project material

1. Introduction

Why the project combines distributed commerce with inventory intelligence

Scalability Bottlenecks

Traditional platforms can place increasing pressure on a single database as users, products and transactions grow. Independent service boundaries provide a path toward targeted scaling.

Real-Time Demands

Online commerce needs fast, secure and reliable experiences with timely inventory visibility. Inventory is therefore treated as a cross-module concern rather than an isolated record.

Inventory Failures

Poor inventory management can lead to stock shortages, overstocking, stale stock information and lost revenue. The platform is designed around clearer stock workflows and synchronization.

Problem Statement & Objectives

Objectives carried forward from the earlier project design

PROBLEM

Build a practical commerce platform that can separate workloads, protect access, manage inventory and support future intelligence without forcing every workload into one storage model.

01

Scalable Distributed Architecture

Services scale independently and communicate through REST APIs.

02

Product & Inventory Management

Track product catalogs, stock levels and warehouse-related data.

03

Secure Customer & Order Management

Authenticated registration/login and an end-to-end order lifecycle.

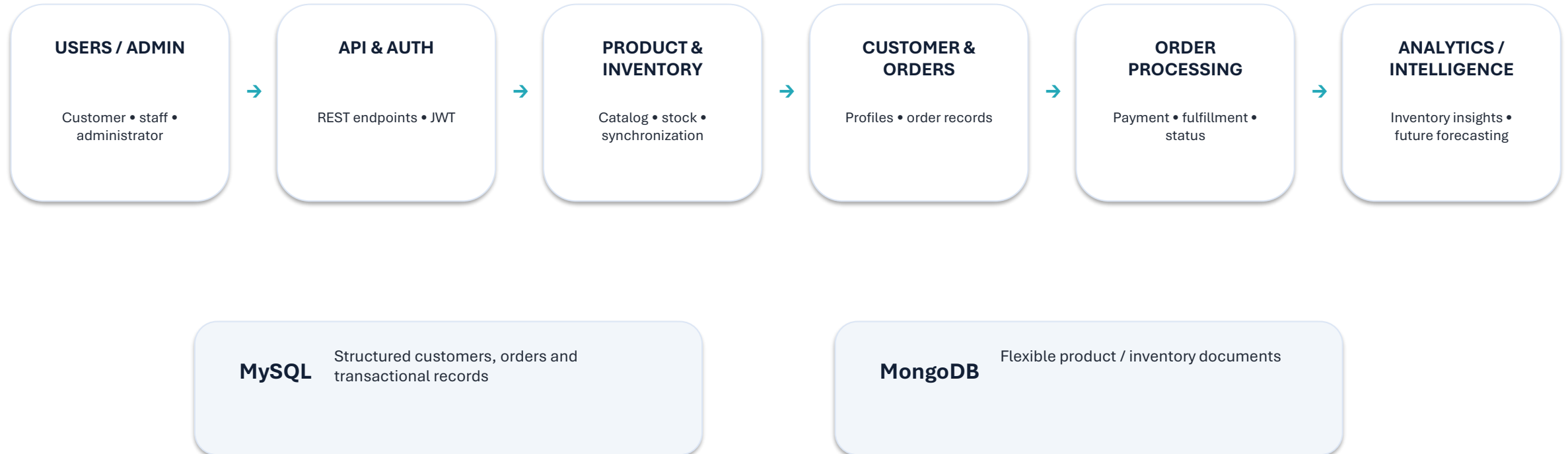
04

Hybrid Database Integration

MySQL for structured transactions and MongoDB for flexible documents.

2. Methodology — System Workflow

Independent services communicate through REST APIs and use suitable persistence



System Architecture

Distributed services + hybrid persistence + API boundary

CLIENTS / ADMIN

Customer • staff • administrator

Authenticated access

API + AUTHENTICATION

REST endpoints • JWT • service boundaries

PRODUCT

Catalog

INVENTORY

Stock / sync

CUSTOMER

Profiles

ORDERS

Order records

PROCESSING

Payment / fulfillment

MYSQL

Structured transactional data

MONGODB

Flexible product / inventory data

Development Methodology

Incremental implementation from architecture → modules → integration → evaluation

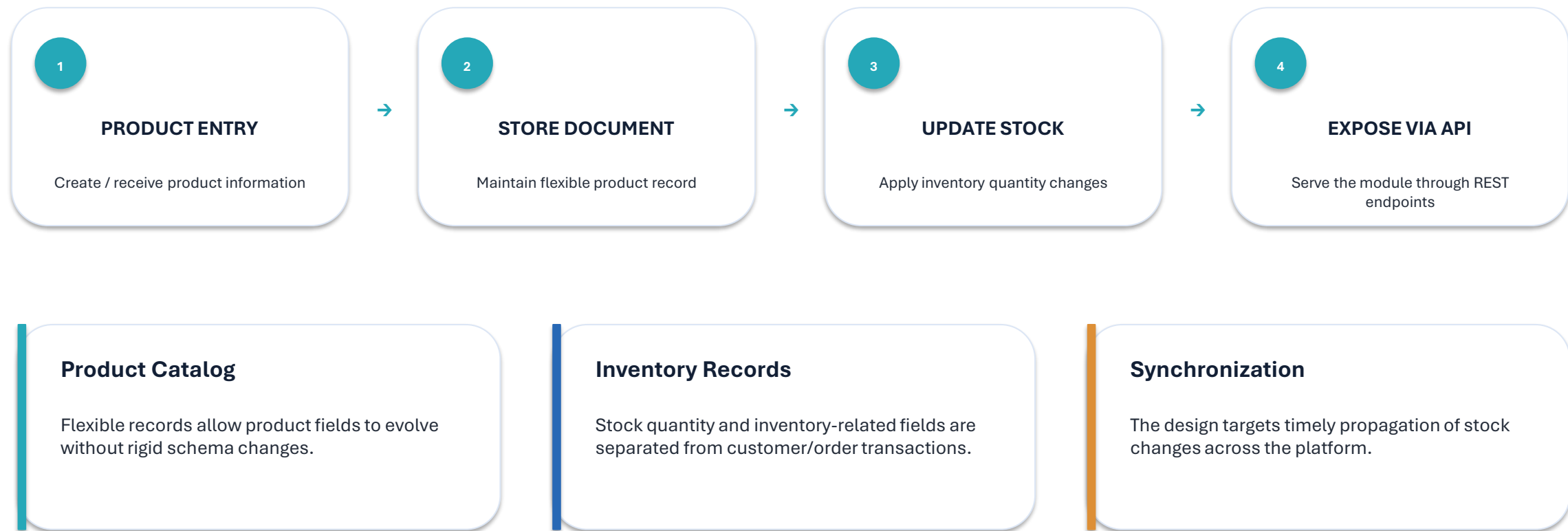


Design principle

Keep services modular so a completed component can be demonstrated and tested without waiting for the entire platform.

11 3. Experiments — Product & Inventory Module

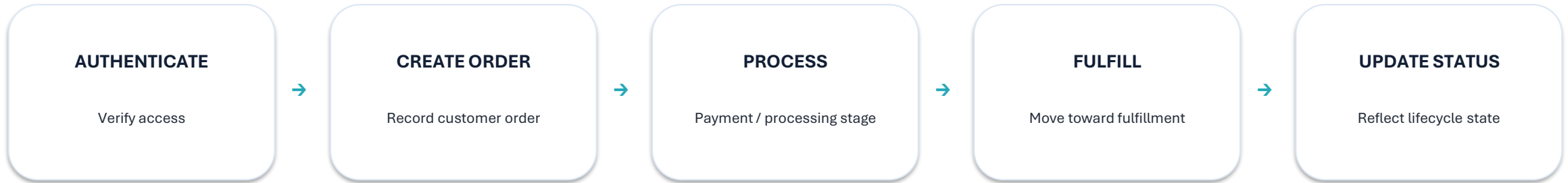
Core catalog and stock workflow supporting inventory intelligence



Current review position: module structure supports separation of product/inventory workloads; synchronization remains an area for continued integration and testing.

12 Experiments — Customer, Orders & Processing

Secure order lifecycle built around authenticated API access



Customer & Authentication

Registration/login and authenticated access form the proposed service boundary.

Order Management

Customer and order records are treated as structured transactional data.

Security Principle

JWT-authenticated REST APIs are intended to verify and protect interactions at the API boundary.

13 Methodology — Hybrid Database Integration

Use each database for the workload it handles best

MYSQL — STRUCTURED & TRANSACTIONAL

- Customer records
- Orders and order lifecycle
- Transaction-oriented relationships
- Consistent structured data

MONGODB — FLEXIBLE DOCUMENTS

- Product catalog
- Inventory-related documents
- Evolving attributes
- Flexible document structure

WHY HYBRID?

Structured transactional workloads and flexible catalog/inventory workloads do not need to be forced into one storage model.

3. Experiments — Functional Test Plan

Evidence expected from the implemented modules

Product

Create / read / update product records

Functional flow and persistence

Inventory

Create / update stock records

Stock change reflected through intended workflow

Orders

Authenticated order lifecycle

Order state changes across processing stages

Integration

API ↔ selected database

Requests reach the intended persistence layer

Security

Protected endpoint access

Unauthorized requests are rejected

Review 3 emphasizes working module evidence rather than final-system claims.

4. Results — Intermediate Implementation Outcomes

Current outcomes reported by the Review 3 material

Product / Inventory

The module structure separates catalog and inventory workloads and supports a flow from product entry to document storage, stock update and API exposure.

Customer / Orders

Customer access, order records and processing are defined around authenticated APIs and a structured transactional workload.

Architecture Outcome

The project has a modular methodology and a clear path from working commerce/inventory modules toward synchronization and inventory intelligence.

Important: the supplied Review 3 deck does not report controlled quantitative benchmark values.

4. Results — Evaluation Framework

Dimensions used to evaluate the current implementation

Functional Correctness

API requests, database operations and workflows produce expected state changes.

Data Consistency

Inventory values should remain accurate during implemented order/update flows.

API & Security

Authentication works and unauthorized requests are rejected.

Modularity

Modules can be developed/tested independently without breaking unrelated services.

Scalability Readiness

Service/database boundaries allow targeted scaling.

Intelligence Readiness

Clean transaction/inventory data can support future forecasting and recommendations.

Quantitative benchmark results are reserved for controlled testing after the current implementation stage.

Novelty & Adaptability

Integration of complementary layers into one practical platform

Integrated Architecture

Combines distributed services, hybrid persistence and commerce workflows rather than solving only one layer.

Real-Time Inventory Focus

Inventory visibility is treated as a cross-module concern to reduce stale-stock risk.

Practical Complexity

Keeps implementation simpler than large multi-model research stacks while leaving room for event streaming.

ADAPTABILITY

Scale services

Add capacity to the service under higher load.

Extend data model

Add product attributes without redesigning every transaction table.

Add intelligence

Use accumulated transaction data for demand forecasting / reorder recommendations.

Add events

Introduce Kafka/CDC or queues for stronger real-time propagation later.

5. Conclusion & Future Work

Current position and path toward final evaluation

CONCLUSION

The project establishes a modular methodology for distributed commerce, hybrid persistence and inventory-focused workflows. The current stage emphasizes implementable service boundaries and testable core modules.

FUTURE WORK

- 1 Complete remaining integrations
- 2 Strengthen inventory synchronization
- 3 Add the intelligence layer
- 4 Benchmark correctness, latency and consistency
- 5 Prepare final evaluation and limitations

Current Limitations & Evaluation Boundary

What the supplied Review 3 material does and does not establish

Implementation Boundary

- Some integrations are explicitly identified as in progress.
- Inventory synchronization still requires stronger connected-workflow testing.
- Final performance claims are intentionally deferred.
- Controlled latency and increasing-load benchmarks are future work.

Evidence Boundary

- No final accuracy/throughput benchmark is reported.
- No final forecasting model results are reported.
- No final comparison against alternative architectures is reported.
- Future event streaming is described as an extension, not a completed component.

References / Source Basis

Project-source material used for this final documentation deck

1

Distributed Digital Commerce and Inventory Intelligence Platform — Review 3 presentation

Primary source for project problem, objectives, methodology, modules, outcomes, evaluation and next steps.

2

PBL Final Documentation Template

Structure used for Abstract, List of Figures, List of Tables, Table of Contents, Introduction, Methodology, Experiments, Results, Conclusion/Future Work and References.

3

Review 2 research grounding carried into Review 3

Review 3 identifies distributed e-commerce, inventory intelligence, hybrid databases and real-time synchronization as the research directions carried forward.

Note: the supplied Review 3 material does not contain a detailed bibliographic reference list; no external citations have been invented.